

Linear Algebra I — Week 2

Matrices over $\mathbb{C}$, Conjugate Transpose, Abstract Spaces & Subspace Tests

Name: _____ Date: _____ All work **by hand** — no calculator, no software.

Key Facts

Complex numbers. $i^2 = -1$; $\overline{a+bi} = a-bi$; $z\bar{z} = |z|^2 = a^2+b^2$; $\frac{z}{w} = \frac{z\bar{w}}{|w|^2}$. Also $\overline{z+w} = \bar{z}+\bar{w}$, $\overline{zw} = \bar{z}\bar{w}$, and $z = \bar{z} \iff z \in \mathbb{R}$.

Matrices over $\mathbb{C}$. $M_{m \times n}(\mathbb{C})$ is the set of $m \times n$ matrices with complex entries. Addition and scalar multiplication are entrywise; $(AB)_{jk} = \sum_l A_{jl}B_{lk}$. Multiplication is associative but *not* commutative.

Conjugate transpose. $(A^*)_{jk} = \overline{A_{kj}}$. Rules: $(A+B)^* = A^*+B^*$, $(\lambda A)^* = \bar{\lambda}A^*$, $(AB)^* = B^*A^*$, $(A^*)^* = A$. A is **Hermitian** if $A^* = A$; **skew-Hermitian** if $A^* = -A$.

Vector space over $\mathbb{F}$. A set V with $+$ and scalar multiplication satisfying: commutativity and associativity of $+$; a zero vector $\mathbf{0}$; additive inverses; $1v = v$; $(\lambda\mu)v = \lambda(\mu v)$; $\lambda(u+v) = \lambda u + \lambda v$; $(\lambda + \mu)v = \lambda v + \mu v$. Standard examples: $\mathbb{F}^n$, $M_{m \times n}(\mathbb{F})$, $P_n(\mathbb{F})$ (polynomials of degree $\leq n$), $\mathcal{F}(S, \mathbb{F})$ (functions $S \rightarrow \mathbb{F}$, pointwise operations).

Subspace test. $W \subseteq V$ is a subspace iff (i) $\mathbf{0} \in W$, (ii) $u, v \in W \Rightarrow u+v \in W$, (iii) $\lambda \in \mathbb{F}, v \in W \Rightarrow \lambda v \in W$. To disprove: exhibit one failure. The field matters: a set can be a subspace over $\mathbb{R}$ but not over $\mathbb{C}$.

Worked Examples

Example 1 · Complex arithmetic

Compute $(2 - i)(1 + 3i)$ and $\frac{5 + i}{2 - i}$.

Solution.

- $(2 - i)(1 + 3i) = 2 + 6i - i - 3i^2 = 5 + 5i$.
- $\frac{5 + i}{2 - i} = \frac{(5 + i)(2 + i)}{(2 - i)(2 + i)} = \frac{10 + 5i + 2i + i^2}{5} = \frac{9 + 7i}{5}$.

Answer: $5 + 5i$ and $\frac{9}{5} + \frac{7}{5}i$

Example 2 · Product over $\mathbb{C}$

Compute $\begin{pmatrix} i & 1 \\ 0 & 2 \end{pmatrix} \begin{pmatrix} 1 & i \\ i & 0 \end{pmatrix}$.

Solution.

- Row 1: $(i \cdot 1 + 1 \cdot i, i \cdot i + 1 \cdot 0) = (2i, -1)$.
- Row 2: $(0 \cdot 1 + 2 \cdot i, 0 \cdot i + 2 \cdot 0) = (2i, 0)$.

Answer: $\begin{pmatrix} 2i & -1 \\ 2i & 0 \end{pmatrix}$

Example 3 · Conjugate transpose

Find M^* for $M = \begin{pmatrix} 2 - i & 3 \\ i & 1 + 2i \end{pmatrix}$.

Solution.

- Transpose: $\begin{pmatrix} 2 - i & i \\ 3 & 1 + 2i \end{pmatrix}$.
- Conjugate every entry.

Answer: $M^* = \begin{pmatrix} 2 + i & -i \\ 3 & 1 - 2i \end{pmatrix}$

Example 4 · Hermitian test

Is $N = \begin{pmatrix} 1 & 2 + i \\ 2 - i & 3 \end{pmatrix}$ Hermitian?

Solution.

- N^* : transpose swaps $2 + i$ and $2 - i$; conjugating gives $\overline{2 - i} = 2 + i$ in position $(1, 2)$ and $\overline{2 + i} = 2 - i$ in position $(2, 1)$.
- Diagonal entries 1, 3 are real and unchanged.

Answer: Yes, $N^* = N$.

Example 5 · Proof: $(AB)^* = B^*A^*$

Prove $(AB)^* = B^*A^*$ for $A \in M_{m \times n}(\mathbb{C})$, $B \in M_{n \times p}(\mathbb{C})$.

Solution.

1. Entry (j, k) of the left side: $\overline{(AB)_{kj}} = \overline{\sum_l A_{kl} B_{lj}} = \sum_l \overline{A_{kl}} \overline{B_{lj}}$ (conjugation respects sums and products).
2. Entry (j, k) of the right side: $\sum_l (B^*)_{jl} (A^*)_{lk} = \sum_l \overline{B_{lj}} \overline{A_{kl}}$.
3. The two sums agree term by term (multiplication in $\mathbb{C}$ commutes).

Answer: $(AB)^* = B^*A^*$.

Example 6 · Axiom failure

On $\mathbb{R}^2$ with usual $\oplus$ and $\lambda \odot (a, b) = (\lambda a, b)$: is it a vector space?

Solution.

1. Test $(\lambda + \mu) \odot v = \lambda \odot v \oplus \mu \odot v$ with $v = (a, b)$.
2. Left: $((\lambda + \mu)a, b)$. Right: $(\lambda a + \mu a, b + b) = ((\lambda + \mu)a, 2b)$.
3. With $b \neq 0$ these differ.

Answer: No: distributivity over scalar addition fails.

Example 7 · Subspace test

Show $W = \{p \in P_2(\mathbb{R}) : p(0) = 0\}$ is a subspace.

Solution.

1. $\mathbf{0}(0) = 0$, so $\mathbf{0} \in W$.
2. $p(0) = q(0) = 0 \Rightarrow (p + q)(0) = 0$.
3. $p(0) = 0 \Rightarrow (\lambda p)(0) = \lambda \cdot 0 = 0$.

Answer: W is a subspace.

Example 8 · Non-subspace

Is $D = \{A \in M_{2 \times 2}(\mathbb{R}) : \det A = 0\}$ a subspace?

Solution.

1. $A = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$ and $B = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$ both have $\det = 0$, so lie in D .
2. $A + B = I$, and $\det I = 1 \neq 0$.

Answer: No: not closed under addition.

Practice Problems

Part A — Complex Arithmetic and Matrices over $\mathbb{C}$

1. [Foundational]
 Compute $(3 + 2i)(1 - 4i)$ and write it in the form $a + bi$.

Step 1. Expand: $(3 + 2i)(1 - 4i) = 3 - 12i + 2i - 8i^2$.

Step 2. Use $i^2 = -1$: $-8i^2 = +8$.

Step 3. Collect: $(3 + 8) + (-12 + 2)i$.

Answer: $11 - 10i$

2. [Foundational]
 Compute $\frac{2 + i}{1 - 3i}$ and write it in the form $a + bi$.

Step 1. Multiply top and bottom by the conjugate $1 + 3i$.

Step 2. Numerator: $(2 + i)(1 + 3i) = 2 + 6i + i + 3i^2 = -1 + 7i$.

Step 3. Denominator: $(1 - 3i)(1 + 3i) = 1 + 9 = 10$.

Answer: $-\frac{1}{10} + \frac{7}{10}i$

3. [Foundational]
 Let $A = \begin{pmatrix} 1 & i \\ 2 - i & 3 \end{pmatrix}$ and $B = \begin{pmatrix} i & 0 \\ 1 & 1 + i \end{pmatrix}$. Compute $A + B$ and $2iA$.

Step 1. Add entrywise: $A + B = \begin{pmatrix} 1 + i & i \\ 3 - i & 4 + i \end{pmatrix}$.

Step 2. Scale each entry of A by $2i$: $2i \cdot 1 = 2i$, $2i \cdot i = -2$, $2i(2 - i) = 4i - 2i^2 = 2 + 4i$, $2i \cdot 3 = 6i$.

Answer: $A + B = \begin{pmatrix} 1 + i & i \\ 3 - i & 4 + i \end{pmatrix}$, $2iA = \begin{pmatrix} 2i & -2 \\ 2 + 4i & 6i \end{pmatrix}$

4. [Intermediate]
 With A and B as in Problem 3, compute AB .

Step 1. $(AB)_{11} = 1 \cdot i + i \cdot 1 = 2i$.

Step 2. $(AB)_{12} = 1 \cdot 0 + i(1 + i) = i + i^2 = -1 + i$.

Step 3. $(AB)_{21} = (2 - i)i + 3 \cdot 1 = 2i - i^2 + 3 = 4 + 2i$.

Step 4. $(AB)_{22} = (2 - i) \cdot 0 + 3(1 + i) = 3 + 3i$.

Answer: $AB = \begin{pmatrix} 2i & -1 + i \\ 4 + 2i & 3 + 3i \end{pmatrix}$

5.

[Foundational]

Find the conjugate transpose C^* of $C = \begin{pmatrix} 1+i & 2 \\ 3i & 4-i \\ 0 & -2i \end{pmatrix}$, and state its size.

Step 1. C is 3×2 , so C^* is 2×3 .

Step 2. Rule: $(C^*)_{jk} = \overline{C_{kj}}$; row j of C^* is the conjugate of column j of C .

Step 3. Column 1 of C is $(1+i, 3i, 0)$, conjugate: $(1-i, -3i, 0)$.

Step 4. Column 2 of C is $(2, 4-i, -2i)$, conjugate: $(2, 4+i, 2i)$.

Answer: $C^* = \begin{pmatrix} 1-i & -3i & 0 \\ 2 & 4+i & 2i \end{pmatrix} \quad (2 \times 3)$

6.

[Intermediate]

Decide whether each matrix is Hermitian ($M^* = M$), skew-Hermitian ($M^* = -M$), or neither: $H = \begin{pmatrix} 2 & 1-i \\ 1+i & 5 \end{pmatrix}$, $K = \begin{pmatrix} i & 1 \\ -1 & 2i \end{pmatrix}$.

Step 1. $H^* = \begin{pmatrix} \overline{2} & \overline{1+i} \\ \overline{1-i} & \overline{5} \end{pmatrix} = \begin{pmatrix} 2 & 1-i \\ 1+i & 5 \end{pmatrix} = H$.

Step 2. $K^* = \begin{pmatrix} \overline{i} & \overline{1} \\ \overline{-1} & \overline{2i} \end{pmatrix} = \begin{pmatrix} -i & -1 \\ 1 & -2i \end{pmatrix}$.

Step 3. $-K = \begin{pmatrix} -i & -1 \\ 1 & -2i \end{pmatrix} = K^*$.

Answer: H is Hermitian; K is skew-Hermitian.

7.

[Intermediate]

Find real numbers x, y such that $M = \begin{pmatrix} 2 & x+iy \\ 3-2i & 5 \end{pmatrix}$ is Hermitian.

Step 1. Hermitian means $M_{21} = \overline{M_{12}}$ (diagonal entries 2, 5 are already real).

Step 2. $3-2i = \overline{x+iy} = x-iy$.

Step 3. Compare real parts: $x = 3$. Compare imaginary parts: $-y = -2$, so $y = 2$.

Answer: $x = 3$, $y = 2$

8.

[Challenge]

Let $A = \begin{pmatrix} 1 & i \\ 2 & 1-i \end{pmatrix}$. Compute $\text{tr}(A^*A)$ without computing all of A^*A , and explain why the result is always a non-negative real number.

Step 1. $A^* = \begin{pmatrix} 1 & 2 \\ -i & 1+i \end{pmatrix}$.

Step 2. Only diagonal entries are needed: $(A^*A)_{11} = 1 \cdot 1 + 2 \cdot 2 = 5$; $(A^*A)_{22} = (-i)(i) + (1+i)(1-i) = 1 + 2 = 3$.

Step 3. $\text{tr}(A^*A) = 5 + 3 = 8$.

Step 4. In general $(A^*A)_{jj} = \sum_k \overline{A_{kj}} A_{kj} = \sum_k |A_{kj}|^2$, so the trace is the sum of $|a_{kj}|^2$ over all entries: here $1 + 1 + 4 + 2 = 8$ ✓.

Answer: 8; it equals $\sum_{j,k} |a_{kj}|^2 \geq 0$.

Part B — Matrix Algebra over $\mathbb{C}$

9.

[Foundational]

Let $A = \begin{pmatrix} 1 & i \\ 0 & 1 \end{pmatrix}$. Compute A^2 and A^3 , then state A^n .

Step 1. $A^2 = \begin{pmatrix} 1 & i \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & i \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 2i \\ 0 & 1 \end{pmatrix}$.

Step 2. $A^3 = A^2 A = \begin{pmatrix} 1 & i+2i \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 3i \\ 0 & 1 \end{pmatrix}$.

Step 3. Pattern (provable by induction): the top-right entry grows by i each time.

Answer: $A^2 = \begin{pmatrix} 1 & 2i \\ 0 & 1 \end{pmatrix}$, $A^3 = \begin{pmatrix} 1 & 3i \\ 0 & 1 \end{pmatrix}$, $A^n = \begin{pmatrix} 1 & ni \\ 0 & 1 \end{pmatrix}$

10.

[Intermediate]

Let $A = \begin{pmatrix} 0 & i \\ 1 & 0 \end{pmatrix}$ and $B = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$. Compute AB , BA and $AB - BA$.

Step 1. $AB = \begin{pmatrix} 0 \cdot 1 + i \cdot 0 & 0 \cdot 0 + i(-1) \\ 1 \cdot 1 + 0 & 0 \end{pmatrix} = \begin{pmatrix} 0 & -i \\ 1 & 0 \end{pmatrix}$.

Step 2. $BA = \begin{pmatrix} 1 \cdot 0 & 1 \cdot i \\ -1 \cdot 1 & 0 \end{pmatrix} = \begin{pmatrix} 0 & i \\ -1 & 0 \end{pmatrix}$.

Step 3. Subtract entrywise.

Answer: $AB - BA = \begin{pmatrix} 0 & -2i \\ 2 & 0 \end{pmatrix} \neq 0$, so $AB \neq BA$.

11.

[Intermediate]

Let $A = \begin{pmatrix} 1 & i \\ 0 & 1 \end{pmatrix}$ and $B = \begin{pmatrix} 1 & 0 \\ i & 1 \end{pmatrix}$. Verify $(AB)^* = B^* A^*$ by computing both sides.

Step 1. $AB = \begin{pmatrix} 1+i \cdot i & i \\ i & 1 \end{pmatrix} = \begin{pmatrix} 0 & i \\ i & 1 \end{pmatrix}$, so $(AB)^* = \begin{pmatrix} 0 & -i \\ -i & 1 \end{pmatrix}$.

Step 2. $A^* = \begin{pmatrix} 1 & 0 \\ -i & 1 \end{pmatrix}$, $B^* = \begin{pmatrix} 1 & -i \\ 0 & 1 \end{pmatrix}$.

Step 3. $B^* A^* = \begin{pmatrix} 1 + (-i)(-i) & -i \\ -i & 1 \end{pmatrix} = \begin{pmatrix} 0 & -i \\ -i & 1 \end{pmatrix}$.

Answer: Both sides equal $\begin{pmatrix} 0 & -i \\ -i & 1 \end{pmatrix}$.

12.

[Challenge]

Let A be an $n \times n$ Hermitian matrix. Prove that (a) every diagonal entry of A is real, and (b) iA is skew-Hermitian.

Step 1. (a) $A^* = A$ gives $\overline{a_{jj}} = a_{jj}$ for each j .

Step 2. A complex number equal to its conjugate is real ($a + bi = a - bi \Rightarrow b = 0$), so $a_{jj} \in \mathbb{R}$.

Step 3. (b) $(iA)^* = \bar{i} A^*$, since conjugate transpose conjugates scalars.

Step 4. $\bar{i} A^* = -i A^* = -i A = -(iA)$, so $(iA)^* = -(iA)$.

Answer: Diagonal entries satisfy $a_{jj} = \overline{a_{jj}}$, hence real; and $(iA)^* = -(iA)$.

Part C — Abstract Vector Spaces

13.

[Intermediate]

On $\mathbb{R}^2$ define $(a, b) \oplus (c, d) = (a + c, b + d)$ and $\lambda \odot (a, b) = (\lambda a, 0)$. Is $(\mathbb{R}^2, \oplus, \odot)$ a vector space over $\mathbb{R}$? Justify.

Step 1. Addition is the usual one, so the additive axioms hold. Check the scalar axiom $1 \odot v = v$.

Step 2. Take $v = (1, 1)$: $1 \odot (1, 1) = (1, 0)$.

Step 3. $(1, 0) \neq (1, 1)$, so the axiom fails.

Answer: No: $1 \odot v \neq v$ for $v = (1, 1)$.

14.

[Challenge]

Let $V = \mathbb{R}_{>0}$ with $x \oplus y = xy$ and $\lambda \odot x = x^\lambda$. (Assume it is a real vector space.) Find the zero vector, the additive inverse of 5, and compute $(3 \odot 2) \oplus ((-1) \odot 5)$.

Step 1. Zero vector $\mathbf{0}$: need $x \oplus \mathbf{0} = x$, i.e. $x \cdot \mathbf{0} = x$, so $\mathbf{0} = 1$.

Step 2. Inverse of 5: need $5 \oplus y = 1$, i.e. $5y = 1$, so $y = \frac{1}{5}$ (also equals $(-1) \odot 5 = 5^{-1}$).

Step 3. $3 \odot 2 = 2^3 = 8$ and $(-1) \odot 5 = 5^{-1} = \frac{1}{5}$.

Step 4. $8 \oplus \frac{1}{5} = 8 \cdot \frac{1}{5} = \frac{8}{5}$.

Answer: $\mathbf{0} = 1$; $-5 = \frac{1}{5}$; result $\frac{8}{5}$.

15.

[Challenge]

Using only the vector space axioms, prove that $0 \cdot v = \mathbf{0}$ for every vector v (here 0 is the scalar zero and $\mathbf{0}$ the zero vector).

Step 1. Since $0 + 0 = 0$ in $\mathbb{F}$: $0 \cdot v = (0 + 0) \cdot v$.

Step 2. Distributivity: $(0 + 0)v = 0v + 0v$, so $0v = 0v + 0v$.

Step 3. Add the additive inverse $-(0v)$ to both sides: $\mathbf{0} = 0v + (-(0v)) = (0v + 0v) + (-(0v))$.

Step 4. Associativity and inverse: the right side is $0v + (0v + (-(0v))) = 0v + \mathbf{0} = 0v$.

Answer: $0 \cdot v = \mathbf{0}$.

16.

[Foundational]

In $P_2(\mathbb{R})$ let $p(x) = 1 + x - x^2$ and $q(x) = 2 - x + x^2$. Compute $3p - 2q$.

Step 1. $3p = 3 + 3x - 3x^2$ and $2q = 4 - 2x + 2x^2$.

Step 2. Subtract coefficientwise: constant $3 - 4 = -1$; x : $3 - (-2) = 5$; x^2 : $-3 - 2 = -5$.

Answer: $-1 + 5x - 5x^2$

17.

[Foundational]

In $\mathcal{F}(\mathbb{R}, \mathbb{R})$ let $f(x) = x^2$ and $g(x) = 3x + 1$. Write the function $2f - g$ and evaluate it at $x = 2$.

Step 1. Functions add and scale pointwise: $(2f - g)(x) = 2x^2 - (3x + 1) = 2x^2 - 3x - 1$.

Step 2. At $x = 2$: $8 - 6 - 1 = 1$.

Answer: $(2f - g)(x) = 2x^2 - 3x - 1$; $(2f - g)(2) = 1$

18.

[Intermediate]

 In $\mathbb{C}^2$ over $\mathbb{C}$, compute $(1+i)(2-i, 3i) + (-i)(1, 1+i)$.

Step 1. $(1+i)(2-i) = 2-i+2i-i^2 = 3+i$ and $(1+i)(3i) = 3i+3i^2 = -3+3i$.

Step 2. $(-i)(1) = -i$ and $(-i)(1+i) = -i-i^2 = 1-i$.

Step 3. Add componentwise: $(3+i-i, -3+3i+1-i)$.

Answer: $(3, -2+2i)$

Part D — Subspace Tests

19.

[Foundational]

 Show that $W = \{(x, y, z) \in \mathbb{R}^3 : x + 2y - z = 0\}$ is a subspace of $\mathbb{R}^3$.

Step 1. Zero vector: $0 + 0 - 0 = 0$, so $(0, 0, 0) \in W$.

Step 2. Addition: if $(x, y, z), (x', y', z') \in W$ then $(x+x') + 2(y+y') - (z+z') = (x+2y-z) + (x'+2y'-z') = 0 + 0 = 0$.

Step 3. Scalars: for $\lambda \in \mathbb{R}$, $\lambda x + 2\lambda y - \lambda z = \lambda(x+2y-z) = 0$.

Step 4. All three conditions hold.

Answer: W is a subspace (contains $\mathbf{0}$, closed under $+$ and scalar multiplication).

20.

[Foundational]

 Show that $W = \{(x, y) \in \mathbb{R}^2 : xy = 0\}$ is *not* a subspace of $\mathbb{R}^2$.

Step 1. $(1, 0) \in W$ since $1 \cdot 0 = 0$, and $(0, 1) \in W$ since $0 \cdot 1 = 0$.

Step 2. Their sum is $(1, 1)$ and $1 \cdot 1 = 1 \neq 0$, so $(1, 1) \notin W$.

Answer: Not closed under addition: $(1, 0) + (0, 1) = (1, 1) \notin W$.

21.

[Challenge]

 Let $W = \{A \in M_{2 \times 2}(\mathbb{C}) : A^* = A\}$ (Hermitian matrices). Is W a subspace of $M_{2 \times 2}(\mathbb{C})$ over $\mathbb{C}$? Is it a subspace over $\mathbb{R}$? Justify both.

Step 1. $\mathbf{0}^* = \mathbf{0}$, so $\mathbf{0} \in W$. For $A, B \in W$: $(A+B)^* = A^* + B^* = A + B$, so W is closed under addition.

Step 2. Real λ : $(\lambda A)^* = \bar{\lambda} A^* = \lambda A$. So W is closed under real scalars.

Step 3. Complex $\lambda = i$: take $A = I \in W$. Then $(iI)^* = -iI \neq iI$, so $iI \notin W$.

Step 4. Hence closure under complex scalar multiplication fails.

Answer: Not a subspace over $\mathbb{C}$ (e.g. $iI \notin W$); it is a subspace over $\mathbb{R}$.

22.

[Intermediate]

 In $P_3(\mathbb{R})$ let $W_1 = \{p : p(2) = 0\}$ and $W_2 = \{p : p(2) = 1\}$. Decide which is a subspace.

Step 1. W_1 : the zero polynomial has value 0 at 2, so $\mathbf{0} \in W_1$.

Step 2. If $p(2) = q(2) = 0$ then $(p+q)(2) = 0$ and $(\lambda p)(2) = \lambda \cdot 0 = 0$; so W_1 is closed.

Step 3. W_2 : the zero polynomial has $\mathbf{0}(2) = 0 \neq 1$, so $\mathbf{0} \notin W_2$.

Answer: W_1 is a subspace; W_2 is not.

23.

[Intermediate]

Show that the set E of even functions $f \in \mathcal{F}(\mathbb{R}, \mathbb{R})$, i.e. $f(-x) = f(x)$ for all x , is a subspace.

Step 1. Zero function: $\mathbf{0}(-x) = 0 = \mathbf{0}(x)$, so $\mathbf{0} \in E$.

Step 2. If $f, g \in E$: $(f + g)(-x) = f(-x) + g(-x) = f(x) + g(x) = (f + g)(x)$.

Step 3. If $f \in E$, $\lambda \in \mathbb{R}$: $(\lambda f)(-x) = \lambda f(-x) = \lambda f(x) = (\lambda f)(x)$.

Answer: E is a subspace of $\mathcal{F}(\mathbb{R}, \mathbb{R})$.

24.

[Challenge]

In $\mathbb{R}^2$ let $U = \{(x, 0)\}$ and $W = \{(0, y)\}$. Show that $U \cap W$ and $U + W$ are subspaces, but $U \cup W$ is not.

Step 1. $U \cap W = \{(0, 0)\}$, the zero subspace.

Step 2. $U + W = \{(x, 0) + (0, y)\} = \{(x, y)\} = \mathbb{R}^2$, a subspace.

Step 3. $(1, 0) \in U \cup W$ and $(0, 1) \in U \cup W$, but $(1, 0) + (0, 1) = (1, 1)$ lies in neither U nor W .

Step 4. So $U \cup W$ is not closed under addition.

Answer: $U \cap W = \{0\}$, $U + W = \mathbb{R}^2$ are subspaces; $U \cup W$ is not.